



# **Engineering Mechanics**

## **Lecture 9: Distributed Force and Friction**

**By**

**Dr. Bhupendra Singh**

**Assistant Professor**

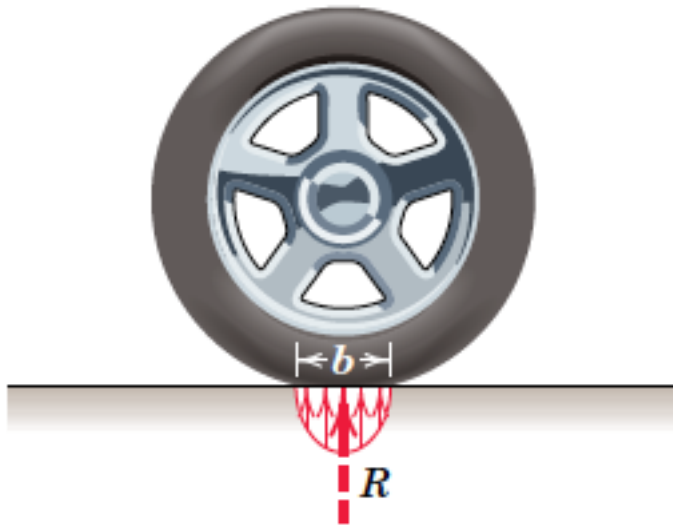
**Civil and Infrastructure Engineering**

**Indian Institute of Technology Jodhpur**

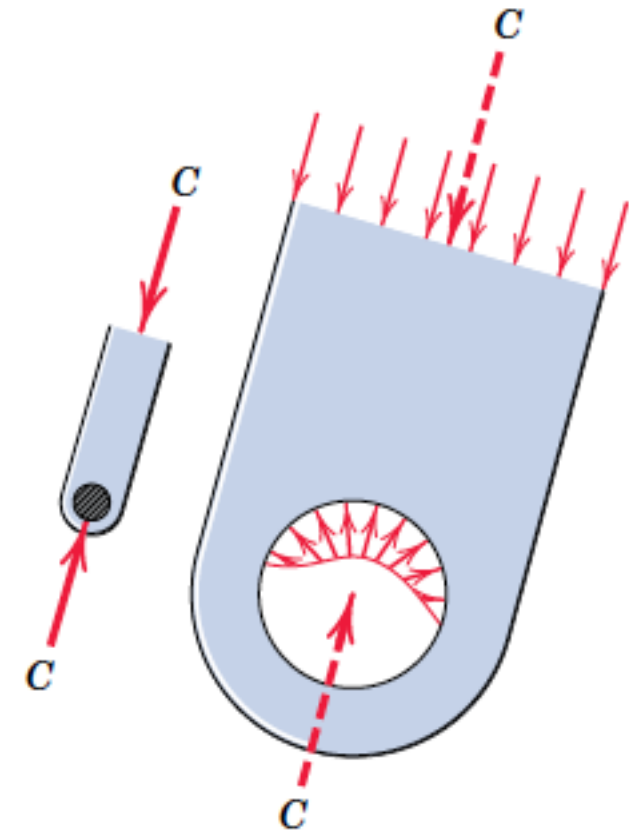
# Distributed forces

- Earlier we treated all forces as *concentrated* along their lines of action and at their points of application.
- This treatment provided a *reasonable model* for those forces.
- Actually, “*concentrated*” forces do not exist in the exact sense
- Every external force applied mechanically to a body is *distributed* over *a finite contact area*, however small.

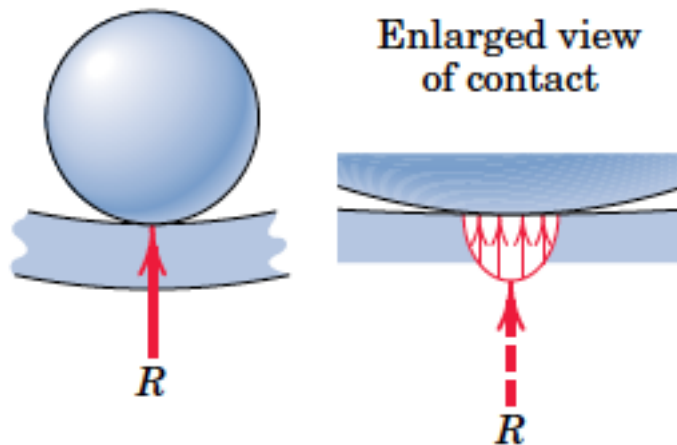
## Distributed forces: Examples



An automobile tire



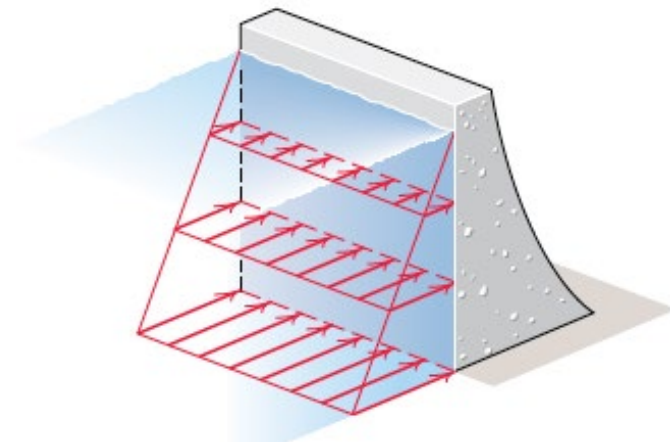
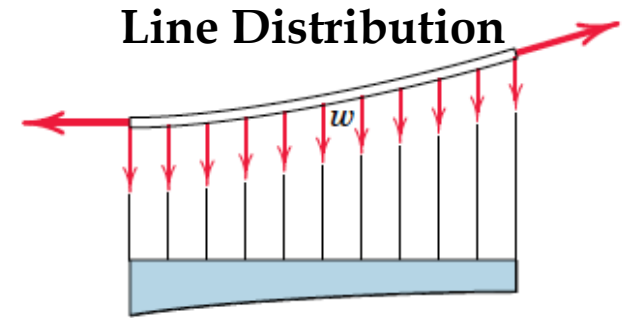
Pin at two-force member of a truss,



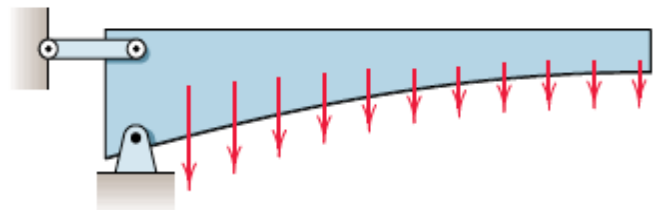
Hardened steel ball and its race in ball bearing

# Type of Distribution

- **Line Distribution:** force is distributed *along a line*
  - Vertical load supported by a suspended cable
  - intensity  $w$  of the loading is expressed as *force per unit length*
- **Area Distribution:** force is distributed *over an area*
  - Hydraulic pressure of water against wall
  - Intensity is expressed as *force per unit area*
- **Volume Distribution:** force is distributed *over the volume* of a body (body force)
  - Gravitational attraction
  - Intensity is expressed as *force per unit volume*



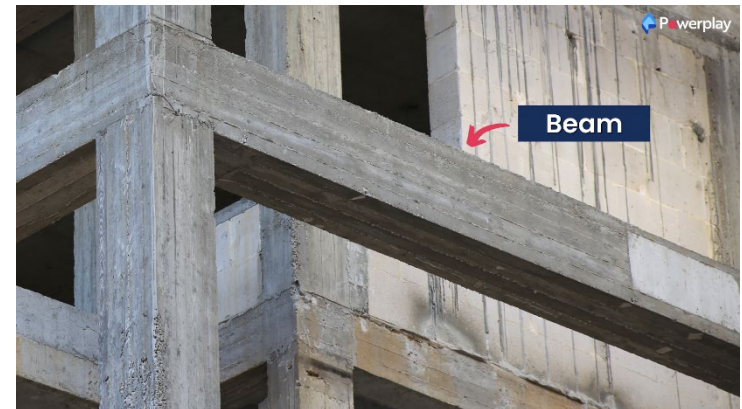
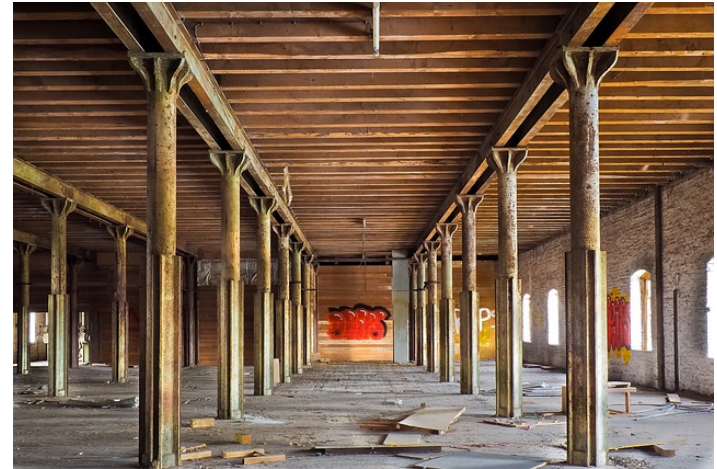
Area Distribution



Volume Distribution

## BEAMS – EXTERNAL EFFECTS

- *Beams* are structural members which offer *resistance to bending* due to applied loads.
- Most beams are *long prismatic bars*, and the *loads* are usually applied *normal to the axes* of the bars.
- Beams are *often subjected to distributed load*.



# Beam: A Prismatic Bar

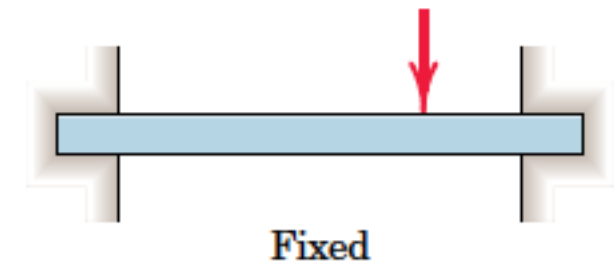
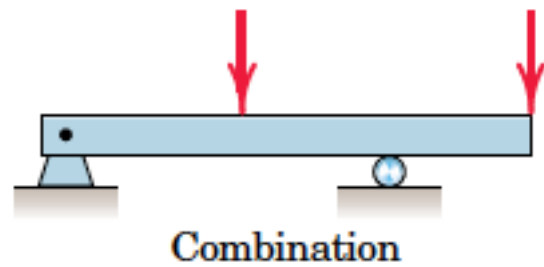
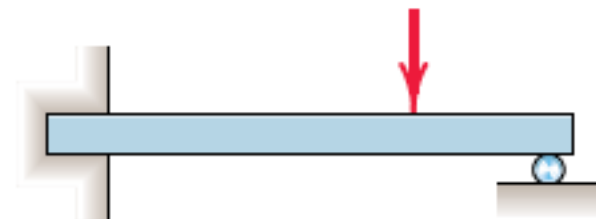
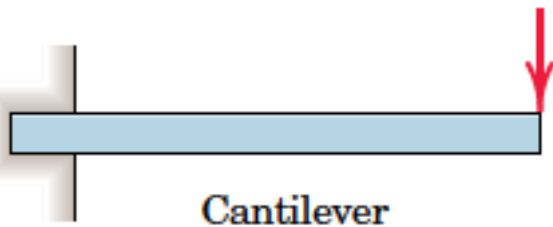
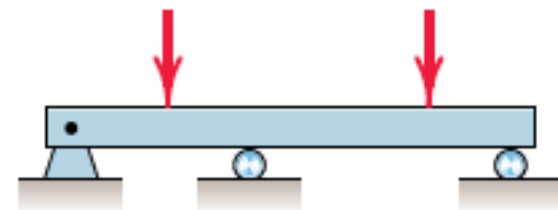
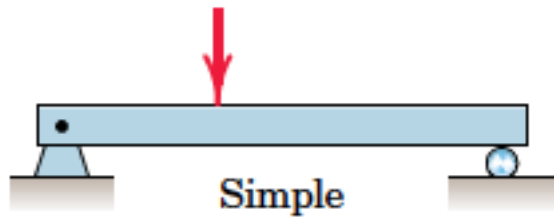


## Prismatic bar

- It means the bar has a uniform cross-section.
- It is a structural member having a straight longitudinal axis and a constant cross-section throughout its length.
- The elongation  $\delta$  of a prismatic bar subjected to a tensile load  $F$  is shown in the figure.
- The elongation is  $\delta = \frac{FL}{AE}$
- The stress at cross-section is given as,  $\sigma = \frac{F}{A}$



# Types of beams



Statically determinate beams

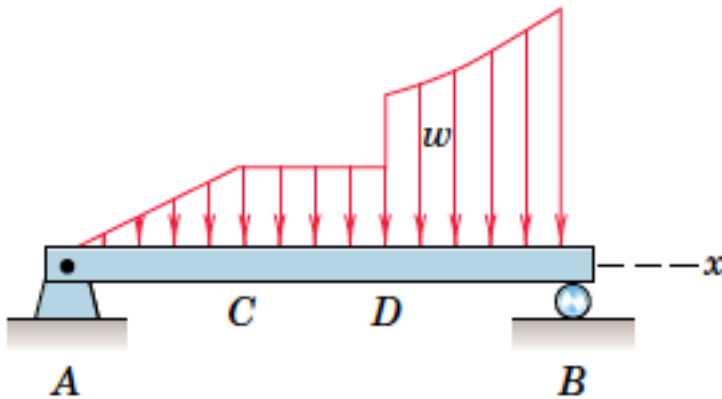
Statically indeterminate beams



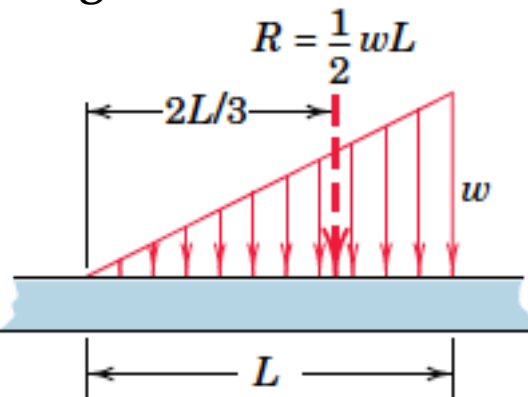
# Beams under Distributed Forces

Solution:

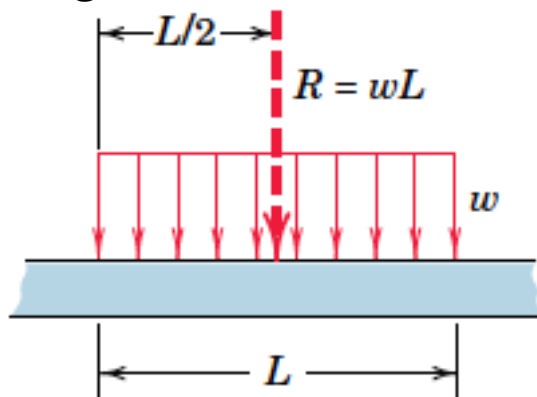
1. Given load intensity is divided into three regions
2. Finally the problem can be solved with three concentrated load **or** with overall load and centroid is obtained using method of Composite Bodies.



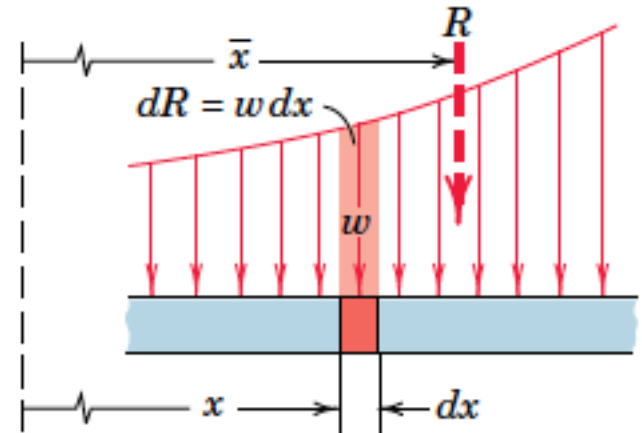
Region 1



Region 2



Region 3



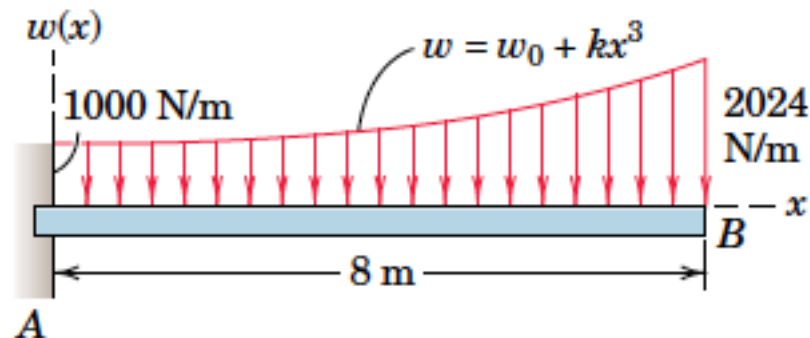
$$dR = w \, dx$$

$$R = \int w \, dx$$

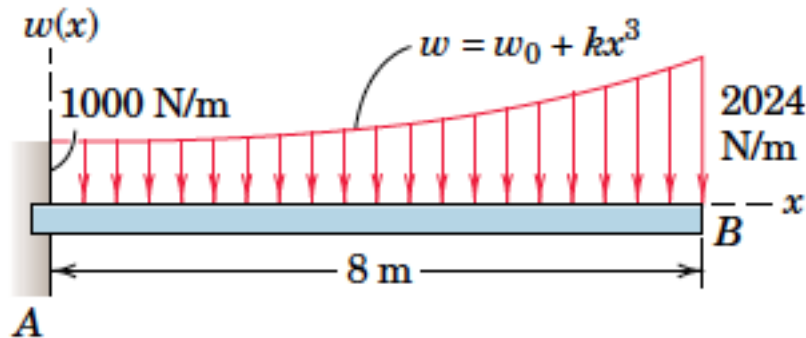
$$\bar{x} = \frac{\int xw \, dx}{R}$$



- Example 1: Determine the reaction at the support A of the loaded cantilever beam.



# Example 1



Using boundary condition

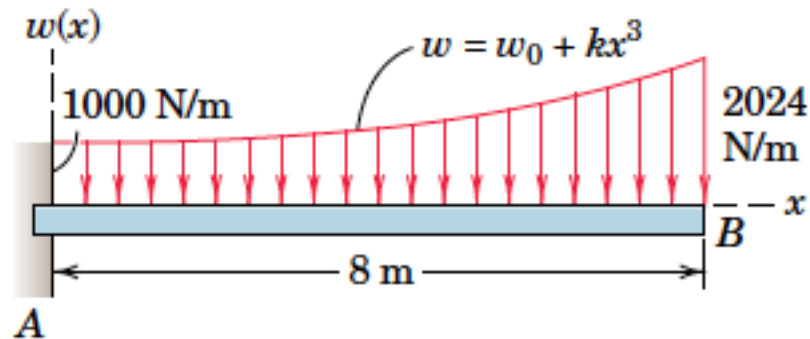
$$w_0 = 1000 \text{ N/m}$$

$$k = 2 \text{ N/m}^4$$

$$R = \int w \, dx = \int_0^8 (1000 + 2x^3) \, dx = \left( 1000x + \frac{x^4}{2} \right) \Big|_0^8 = 10 \, 050 \text{ N}$$

$$\bar{x} = \frac{\int xw \, dx}{R} = \frac{1}{10 \, 050} \int_0^8 x(1000 + 2x^3) \, dx = \frac{1}{10 \, 050} (500x^2 + \frac{2}{5}x^5) \Big|_0^8 = 4.49 \text{ m}$$

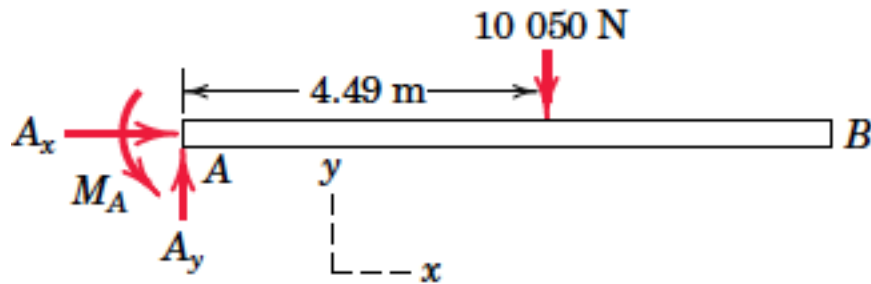
# Example 1



$$[\Sigma M_A = 0]$$

$$M_A - (10\,050)(4.49) = 0$$

$$M_A = 45\,100 \text{ N}\cdot\text{m}$$



$$[\Sigma F_y = 0]$$

$$A_y = 10\,050 \text{ N}$$

$$A_x = 0$$

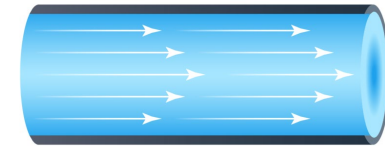
# Friction

- So far we have assumed that the *forces of action and reaction* between contacting surfaces act *normal to the surfaces*.
- Many problems in which we must consider the ability of contacting surfaces to support *tangential* as well as *normal* forces.
- *Tangential forces* generated between contacting surfaces are called *friction forces* and occur to some degree in the interaction between all real surfaces.

## Examples

### Friction as disadvantage

- Bearings of all types, power screws, gears,
- the flow of fluids in pipes, and the propulsion of aircraft and missiles through the atmosphere



### Friction as advantage

- brakes, clutches, belt, drives, and wedges



# Types of Friction

- Dry Friction
- Fluid Friction
- Internal Friction

## Types of Friction

- *Dry Friction*

- Dry friction occurs when the *unlubricated surfaces of two solids are in contact* under a condition of sliding or a tendency to slide
- A friction force *tangent to the surfaces* of contact
- The direction of this friction force always *opposes the motion* or impending motion.
- This type of friction is also called *Coulomb friction*.



# Types of Friction

- *Fluid Friction*

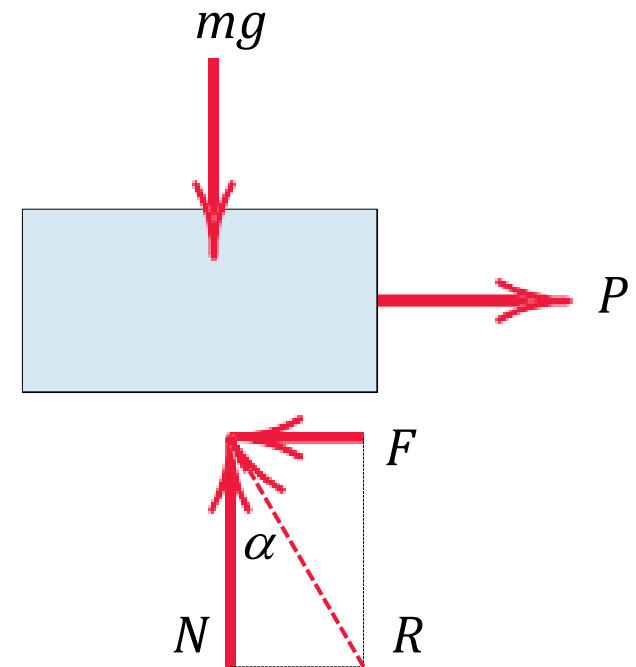
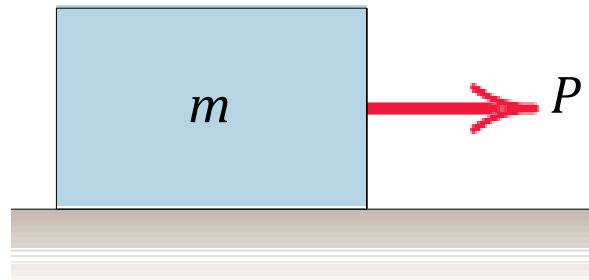
- It occurs when *adjacent layers in a fluid* (liquid or gas) are *moving at different velocities*.
- This motion causes *frictional forces* between fluid elements, and these forces depend on the relative velocity between layers.
- When there is *no relative velocity*, there is *no fluid friction*.
- Fluid friction also depends on *the viscosity of the fluid*, which is a measure of its resistance to shearing action between fluid

## Types of Friction

- *Internal Friction*

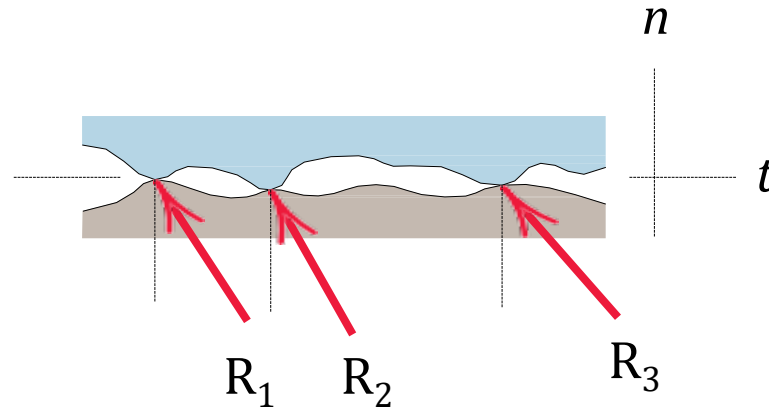
- Internal friction occurs in *all solid materials* which are subjected to *cyclical loading*.
- For *highly elastic materials* the recovery from deformation occurs with very little loss of energy due to *low internal friction*.

# Mechanism of Dry Friction



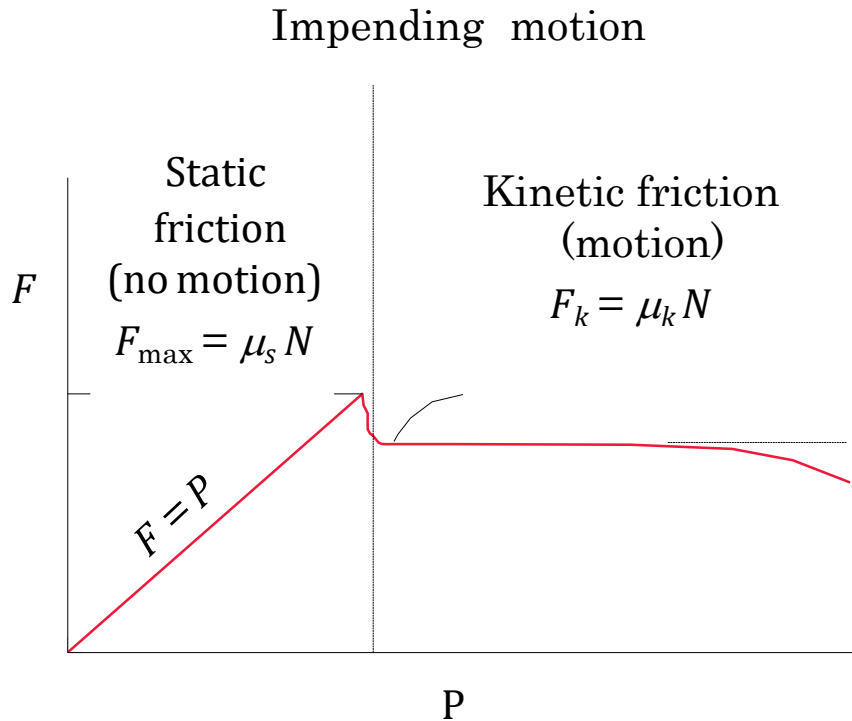
- $P$  continuously increases from zero to a value sufficient to move the block and give it an appreciable velocity.

# Irregularities of the mating surfaces



- The *total normal force  $N$*  is the sum of the  *$n$ -components* of the  $R$ 's, and the *total frictional force  $F$*  is the sum of the  *$t$ -components* of the  $R$ 's.
- Friction not only depend on *geometric profile* of the irregularities but also on the *extent of local deformation* at each contact point.

# Static Friction



Motion is impending

$$F_{\max} = \mu_s N$$

Condition of static equilibrium when motion is not impending

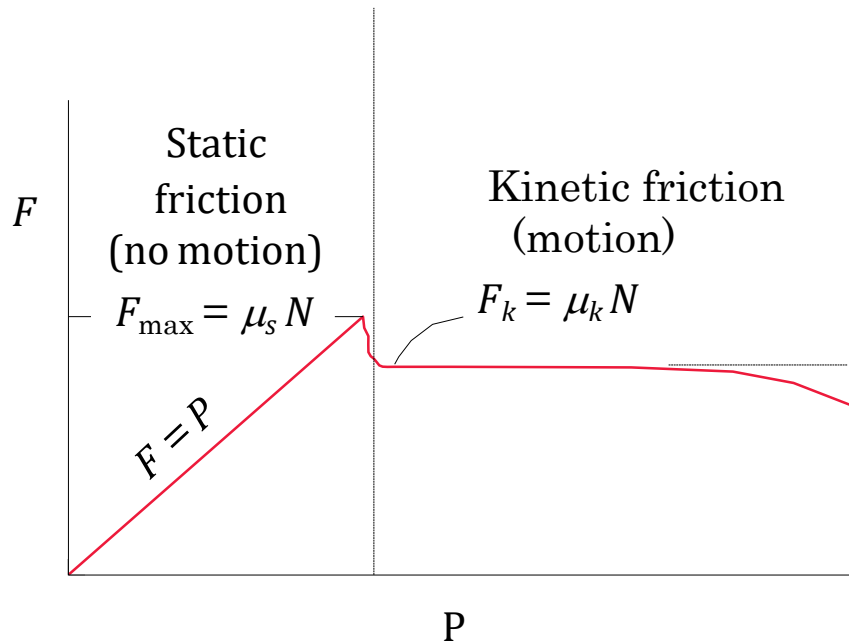
$$F < \mu_s N$$

$\mu_s$  is the coefficient of static friction

- The region in to the point of slippage or impending motion is called *the range of static friction*,
- In this range the value of the friction force is determined by the *equations of equilibrium*

# Kinetic Friction

Impending motion



In Motion

$$F_k = \mu_k N$$

$\mu_k$  is the coefficient of kinetic friction

- After slippage occurs, a *condition of kinetic friction* accompanies the ensuing motion.
- Kinetic friction force is *usually somewhat less than* the maximum static friction force

# Friction Angles

The direction of the resultant  $R$  measured from the direction of  $N$  is specified by

$$\tan \alpha = F / N$$

When the friction force reaches its limiting static value  $F_{max}$ , the angle  $\alpha$  reaches a maximum value  $\phi_s$ . Thus,

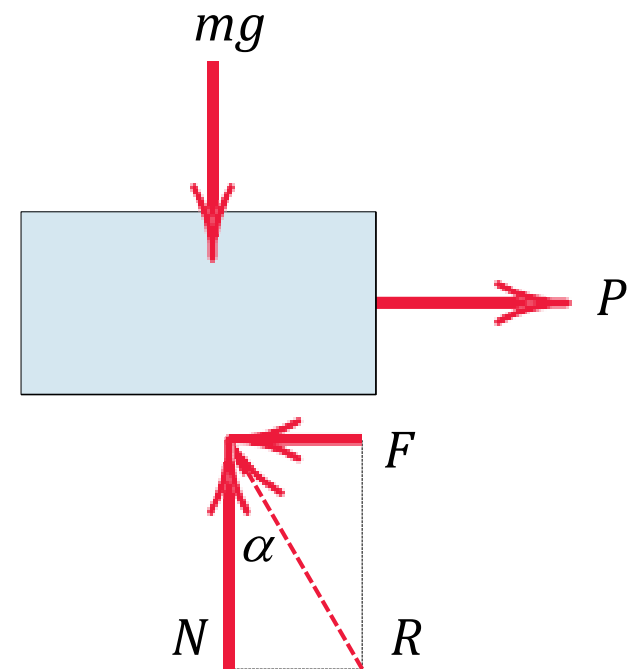
$$\tan \phi_s = \mu_s$$

When slippage is occurring, the angle  $\alpha$  has a value  $\phi_k$

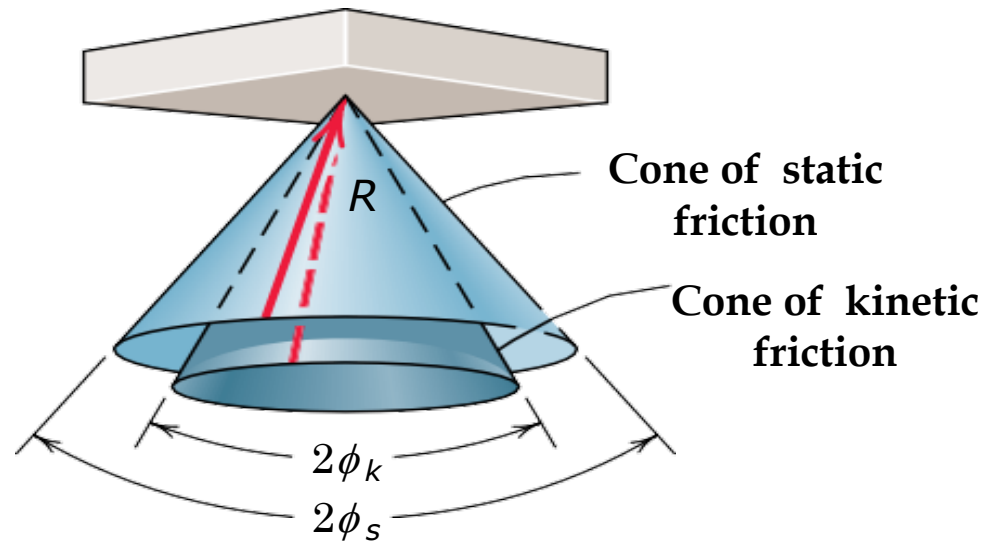
$$\tan \phi_k = \mu_k$$

In practice we often see the expression

$$\tan \phi = \mu$$



# Friction cone



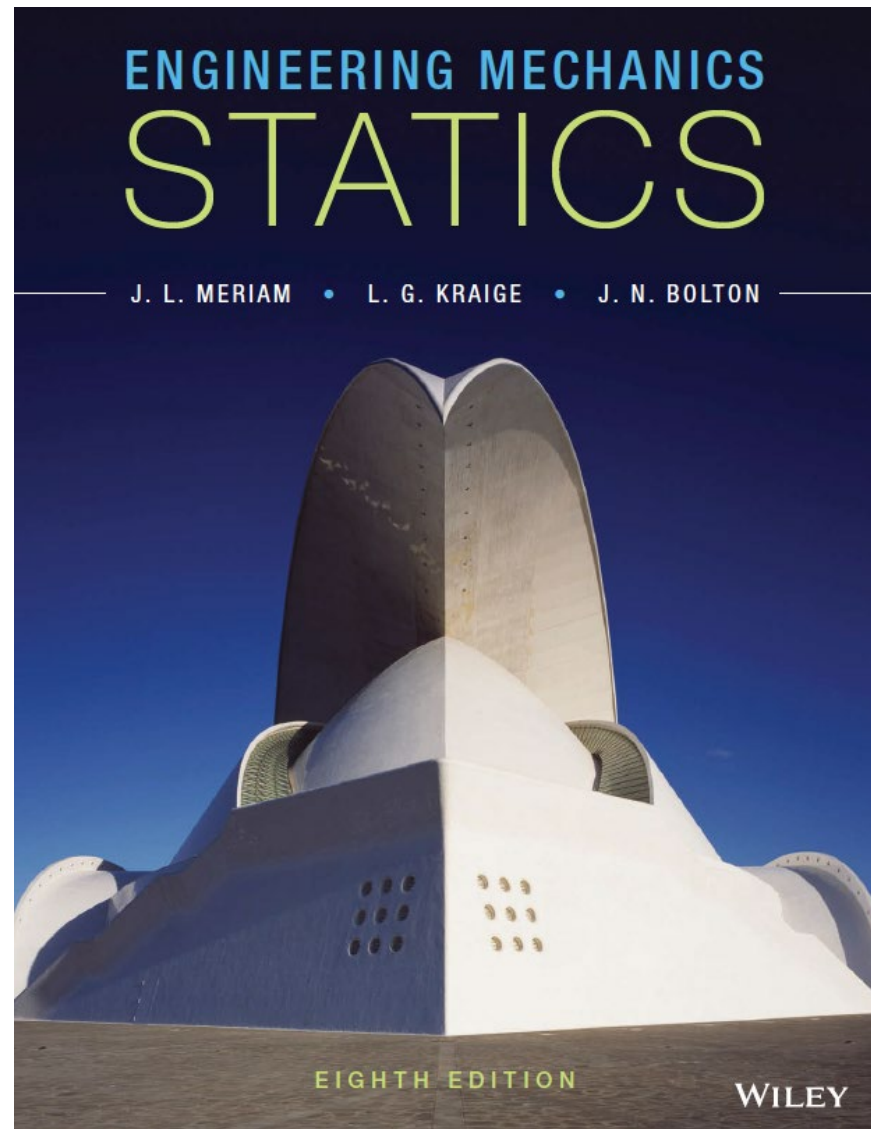
- **Motion is impending:**  $R$  must be *one element* of a right-circular cone of vertex angle  $2\phi_s$ ,
- **Motion is not impending:**  $R$  is *within* the cone. This cone of vertex angle  $2\phi_s$  is called *the cone of static friction*
- **Motion occurs:** the angle of kinetic friction applies, and the reaction must lie on the surface of a slightly different cone of vertex angle  $2\phi_k$ . This cone is the *cone of kinetic friction*.



## Type of problems

- Case 1: Body which is in equilibrium is on the verge of slipping
  - Friction force:  $F_{max} = \mu_s N$
- Case 2: relative motion is known to exist
  - Friction force:  $F_k = \mu_k N$
- Case 3: neither the condition of impending motion nor the condition of motion is known
  - Under Static equilibrium: Friction force:  $F < (F_{max} = \mu_s N)$
  - On the verge of slipping : Friction force:  $F = (F_{max} = \mu_s N)$
  - In motion:  $F > (F_{max} = \mu_s N)$ , Friction force:  $F_k = \mu_k N$

## References :



# THANK YOU